



UNSW
SYDNEY

MATH1231 Mathematics 1B – Algebra

Section 3: Subspaces

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Slides: David Angell, Laure Helme-Guizon and many others



Definition. A **vector space** over a field $\mathbb{F}$ is a non-empty set V of vectors, together with an operation of vector addition and an operation of multiplication by scalars from $\mathbb{F}$, for which the following properties hold.

1. **Closure under (vector) addition.** If $\vec{u}, \vec{v}$ are in V , then $\vec{u} + \vec{v}$ is in V .
2. **Associativity of (vector) addition.** For all $\vec{u}, \vec{v}, \vec{w}$ in V we have $(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$.
3. **Commutativity of (vector) addition.** For all $\vec{u}, \vec{v}$ in V we have $\vec{u} + \vec{v} = \vec{v} + \vec{u}$.
4. **Zero vector.** There exists an element $\vec{0}$ of V such that for all $\vec{v} \in V$ we have $\vec{v} + \vec{0} = \vec{v}$.
5. **Negatives.** For each element $\vec{v}$ of V there is an element $\vec{w}$ of V such that $\vec{v} + \vec{w} = \vec{0}$. (We usually denote this element by $-\vec{v}$.)
6. **Closure under scalar multiplication.** If λ is in $\mathbb{F}$ and $\vec{v}$ is in V , then $\lambda \vec{v}$ is in V .
7. **"Associativity" of multiplication.** For all $\lambda, \mu \in \mathbb{F}$ and all $\vec{v} \in V$ we have $\lambda(\mu \vec{v}) = (\lambda\mu) \vec{v}$.
8. **Multiplication by (the scalar) 1.** For each $\vec{v} \in V$ we have $1 \vec{v} = \vec{v}$.
9. **Scalar distributive law.** For all λ, μ in $\mathbb{F}$ and all $\vec{v}$ in V we have $(\lambda + \mu) \vec{v} = \lambda \vec{v} + \mu \vec{v}$.
10. **Vector distributive law.** For all λ in $\mathbb{F}$ and all $\vec{u}, \vec{v}$ in V we have $\lambda(\vec{u} + \vec{v}) = \lambda \vec{u} + \lambda \vec{v}$.

Today's big question :

How to tell if a set is a vector space or not?



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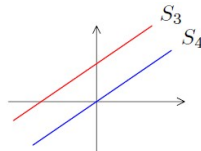
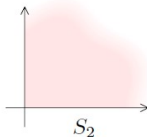
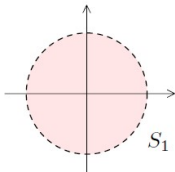
Let's start with the easier one, i.e how to prove a set is / is not (*select one*) a vector space.

To prove a set _____ a vector space, we only need to show that _____

Today's big question #1 :

How do we prove something is NOT a vector space?

Example 1. Which of S_1, S_2, S_3, S_4 are vector spaces?



- S_1 is not a vector space because it is not closed under addition.
- S_2 is closed under addition, but is not closed under (scalar) multiplication, so it is not a vector space.
- S_3 is not a vector space because it fails both of these tests.

Recall terminology from the vector space axioms: “ S is closed under addition” means

$$\text{if } \vec{u} \in S \text{ and } \vec{v} \in S \text{ then } \vec{u} + \vec{v} \in S ,$$

while “ S is closed under scalar multiplication” means

$$\text{if } \lambda \text{ is a scalar and } \vec{v} \in S \text{ then } \lambda \vec{v} \in S .$$

Ways to show that a given set is **not** a vector space.

TIP!

It is sometimes very easy to show that a given set is **not** a vector space.

- If S does not contain the zero vector, then S is not a vector space.
- If we can find a vector $\vec{v}$ in S such that $-\vec{v}$ is not in S , then S is not a vector space.

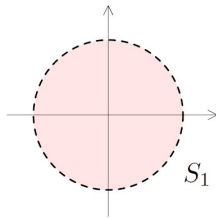
Examples 2.

1. Let $S = \{A \in M_{3,3}(\mathbb{R}) : A^{-1} \text{ exists}\}$. Then S does not contain the zero matrix, so S is not a vector space.
2. Consider again $S_2 = \{\text{vectors in } \mathbb{R}^2 \text{ pointing into the first quadrant}\}$. Then, for example, $(1, 1)^T$ is in S_2 , but $-(1, 1)^T = (-1, -1)^T$ which is not in S_2 . Therefore S_2 is not a vector space.
3. Let $S = \{p \in \mathbb{P}(\mathbb{R}) : p(2) \geq 0\}$. Take, for example, $p(x) = x^2$. Then
 - p is in S because $p(2) = 4 \geq 0$; but
 - $-p$ is not in S because $-p(2) = -4 \not\geq 0$.Therefore S is not a vector space.

Ways to show that a given set is **not** a vector space.

Comments.

- In showing that a set need not contain negatives, you should always give a *specific numerical example*. You should do the same when showing that a set is not closed under addition, or not closed under scalar multiplication.
- These two methods **do not** work in reverse. For example, S_1 , the set of vectors of length less than a given value *does* contain the zero vector; and if $\vec{v}$ is in S_1 then $-\vec{v}$ is *always* in S_1 ; but still, S_1 is not a vector space.



In other words: you can (sometimes) use these methods to show that something is not a vector space, but you can **never** use them to show that something is a vector space.

Today's big question #2 :

How do we prove something IS a vector space?



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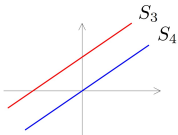
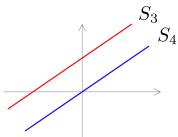
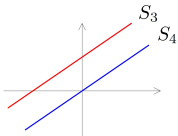
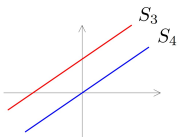
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Checking the 10 axioms one by one will always work, but could there be a shortcut in some cases?

Let's investigate and **DISCOVER** a shortcut.

Example 3. Show that the line S_4 is a vector space.



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If S is a subset of a vector space, some axioms must be true.

Note again that in order to prove that S_4 was a vector space, we did not really need to check axioms 2, 3, 7, 8, 9 or 10. The reason for this was that

- S_4 is a subset of $\mathbb{R}^2$; and
- we already know that $\mathbb{R}^2$ is a vector space.

So it was only necessary to check axioms 1, 4, 5 and 6.

It gets even better:

In fact, if we have proved that axioms 1, 4 and 6 are true, then axiom 5 also works automatically and need not be checked separately.

Reason: we have checked axiom 6, so we know that

$$\text{if } \lambda \text{ is any scalar and } \vec{v} \in S_4 \text{ then } \lambda \vec{v} \in S_4 .$$

In particular, this is true for the scalar $\lambda = -1$, that is,

$$\text{if } \vec{v} \in S_4 \text{ then } -\vec{v} \in S_4 ,$$

which shows that the negative axiom (axiom 5) is true.

Subspace of a vector space

Definition (subspace)

If V is a vector space, and S is a subset of V , and S itself is also a vector space with the same scalar field, the same operation of addition and the same operation of scalar multiplication as V , then S is called a **subspace** of V .



Make sure you understand the difference between the words *subset* and *subspace*.

Example 4. The set $S_5 = \{\lambda(2\vec{i} - \vec{j}) : \lambda \in \mathbb{R}\}$ is a subspace of the vector space $\mathbb{R}^2$ (it is also a subset of the set $\mathbb{R}^2$).

'The subspace theorem' will soon give us a way to easily prove that S_5 is a subspace of the vector space $\mathbb{R}^2$. For now, just believe me. I would just like you to have a concrete example to cling to so you can start forming your intuitive mental model of a subspace.

How to prove that a given subset of a vector space is itself a vector space?

What we've just discussed shows that if we know that V is a vector space and S is a subset of V , and if we want to prove that S is also a vector space, there are just three things to check.

Theorem (The Subspace Theorem)

Let V be a vector space and S a subset of V . If the following conditions are (all) true,

- $\vec{0}$ is in S ;
- S is closed under addition;
- S is closed under scalar multiplication;

then S is a subspace of V (and so S is a vector space).

TIP!

In practice, the Subspace Theorem gives the simplest way to prove that a set S forms a vector space. We shall keep in mind a number of "standard" vector spaces ($\mathbb{R}^n$, $M_{m,n}$, {polynomials} and so on), identify S as a subset of one of these, and check the three necessary conditions.

How to use the Subspace theorem: Subspace of what?

In order to use the subset theorem to prove that some set S is a vector space, you will first need to identify a vector space containing your set S .



This is where to look: In all the sets below, m and n are positive integers and $\mathbb{F}$ is a field. Usually, $\mathbb{F}$ is $\mathbb{Q}$, $\mathbb{R}$ or $\mathbb{C}$.

- $\mathbb{R}^n$ is a vector space over the field $\mathbb{R}$.
- $\mathbb{C}^n$ is a vector space over the field $\mathbb{C}$ (and also over the field $\mathbb{R}$).
- $\mathbb{P}(\mathbb{F})$, the set of polynomials with coefficients in the field $\mathbb{F}$, is a vector space over $\mathbb{F}$.
- $\mathbb{P}_n(\mathbb{F})$, the set of polynomials of degree at most n with coefficients in the field $\mathbb{F}$, is a vector space over $\mathbb{F}$.
- $M_{mn}(\mathbb{F})$, the set of matrices with m rows and n columns and with entries in the field $\mathbb{F}$, is a vector space over $\mathbb{F}$.

Homework

Note/exercise/homework.

The first important thing in this subject is to make sure you know what all the words mean! So far, you should know *precisely* what is meant by the terms vector space; closed under addition; closed under scalar multiplication; subspace.

SUMMARY.

■ To prove that a given subset of a vector space *is* itself a vector space, use :

Theorem (The Subspace Theorem)

Let V be a vector space and S a subset of V . If the following conditions are (all) true,

- $\vec{0}$ is in S ;
- S is closed under addition;
- S is closed under scalar multiplication;

then S is a subspace of V (and so S is a vector space).

■ To show that a subset S of a vector space is **not** a vector space, it is enough

- to show that S does not contain the zero vector,
or
- to show that S is not closed under addition,
or
- to show that S is not closed under (scalar) multiplication, which can often be done by showing that there exist a vector $\vec{v}$ in S such that $-\vec{v}$ in S .

Subspaces of $\mathbb{R}^n$: Examples

We used geometric arguments to show that S_4 is a vector space. Now let's do the same thing using algebra.

Exercise 5. Show that $S_4 = \{\vec{x} \in \mathbb{R}^2 : 2x_1 - 3x_2 = 0\}$ is a vector space.



Subspaces of $\mathbb{R}^n$: Examples

Exercise 6. Show that $S_5 = \{\mathbf{x} \in \mathbb{R}^3 : 4x_1 - x_2 - 7x_3 = 0\}$ is a vector space.

Similar to S_4 , left as an exercise.



Subspaces of $\mathbb{R}^n$: Examples

Exercise 7. Is $S_6 = \{\vec{x} \in \mathbb{R}^3 : 4x_1 - x_2 - 7x_3 = -5\}$ a vector space?



Subspaces of $\mathbb{R}^n$

What do our recent examples mean geometrically? Well, S_5 is a plane through the origin in $\mathbb{R}^3$, while S_6 is a plane not through the origin. Similarly, S_4 is a line through the origin in $\mathbb{R}^2$, whereas S_3 is a line not through the origin. The sets S_1 and S_2 are not lines or planes at all.

It can be shown that the subspaces of $\mathbb{R}^2$ are

- a single point, $\{\vec{0}\}$;
- any line through the origin;
- the whole plane $\mathbb{R}^2$.

The subspaces of $\mathbb{R}^3$ are

- a single point, $\{\vec{0}\}$;
- any line through the origin;
- any plane through the origin;
- the whole of 3-dimensional space, $\mathbb{R}^3$.

Examples beyond $\mathbb{R}^n$: when 'vectors' are polynomials.

Exercise 8. Let S be the set of polynomials which satisfy all of the following conditions:

- their coefficients are complex numbers,
- their degree is at most 7,
- their second derivative evaluated at $x = 5$ is zero.

a) Rewrite S using set notations.

b) Is S a vector space over the field $\mathbb{C}$?



Examples beyond $\mathbb{R}^n$: when 'vectors' are polynomials.

Exercise 8, continued. Let S be the set of polynomials which satisfy all of the following conditions:

- their coefficients are complex numbers,
- their degree is at most 7,
- their second derivative evaluated at $x = 5$ is zero.

a) Rewrite S using set notations.

b) Is S a vector space over $\mathbb{C}$?



Examples beyond $\mathbb{R}^n$: when 'vectors' are polynomials.

Exercise 9. Consider the set

$$S = \{P \in \mathbb{P}_{16}(\mathbb{R}) : P'(9) = -3\}$$

Is S a vector space?



A poorly written proof

Exercise 10. Criticise the following vector space proof. Find all the mistakes, and also try to identify any parts which are not exactly wrong but could have been expressed better.

Problem. Show that $S = \{ \vec{x} \in \mathbb{R}^4 \mid x_1 - 6x_2 + x_4 = 0 \}$ is a vector space.

Proof. Clearly S contains 0.

Let $\vec{x}, \vec{y} \in \mathbb{R}^4$. Then

$$x_1 - 6x_2 + x_4 = 0 \quad y_1 - 6y_2 + y_4 = 0 ,$$

$$\begin{aligned} \text{so} \quad \vec{x} + \vec{y} &= (x_1 - 6x_2 + x_4) + (y_1 - 6y_2 + y_4) \\ &= 0 + 0 \\ &= 0. \end{aligned}$$

So $\vec{x}$ is closed under addition.

Let $\lambda \in \mathbb{R}$. Then

$$x_1 - 6x_2 + x_3 = 0.$$

Therefore

$$(\lambda x_1) - 6(\lambda x_2) + (\lambda x_3) = \lambda(x_1 - 6x_2 + x_3) = \lambda 0 = 0$$

so it is closed under multiplication.

A well written proof

Example 11. Show that $S = \{(x_1, 2x_2, x_2, x_1 + x_2) \mid x_1, x_2 \in \mathbb{R}\}$ is a vector space over $\mathbb{R}$.

Worked solution.

1. S is a subset of $\mathbb{R}^4$, which we know is a vector space.
2. By taking $x_1 = x_2 = 0$ we see that S contains the zero vector $(0, 0, 0, 0)$.
3. Let $\vec{x}$ and $\vec{y}$ be in S . Then

$$\vec{x} = (x_1, 2x_2, x_2, x_1 + x_2) \quad \vec{y} = (y_1, 2y_2, y_2, y_1 + y_2)$$

for some $x_1, x_2, y_1, y_2 \in \mathbb{R}$. Therefore

$$\begin{aligned}\vec{x} + \vec{y} &= (x_1 + y_1, 2x_2 + 2y_2, x_2 + y_2, (x_1 + x_2) + (y_1 + y_2)) \\ &= (x_1 + y_1, 2(x_2 + y_2), x_2 + y_2, (x_1 + y_1) + (x_2 + y_2)) \\ &= (z_1, 2z_2, z_2, z_1 + z_2)\end{aligned}$$

where $z_1 = x_1 + y_1$ and $z_2 = x_2 + y_2$. Since z_1 and z_2 are real numbers, $\vec{x} + \vec{y}$ is in S . Therefore S is closed under addition.

Example 11, continued

4. Let $\vec{x} \in S$ and $\lambda \in \mathbb{R}$. Then

$$\vec{x} = (x_1, 2x_2, x_2, x_1 + x_2)$$

for some real numbers x_1 and x_2 . Therefore

$$\begin{aligned}\lambda \vec{x} &= (\lambda x_1, \lambda(2x_2), \lambda x_2, \lambda(x_1 + x_2)) \\ &= (\lambda x_1, 2(\lambda x_2), \lambda x_2, (\lambda x_1) + (\lambda x_2)) \\ &= (w_1, 2w_2, w_2, w_1 + w_2)\end{aligned}$$

where $w_1 = \lambda x_1$ and $w_2 = \lambda x_2$ are real numbers, and so $\lambda \vec{x}$ is in S . Hence S is closed under scalar multiplication.

By the Subspace Theorem, S is a subspace of $\mathbb{R}^4$, and so S is a vector space.

Comment. Once again, be careful with writing out your proof – the “actual mathematics” is not the only important thing! Observe that in step 4, some of the brackets we have written are not algebraically necessary; but they help to make our meaning clear and thereby improve our argument.

A past exam question. (time permitting)

For each of the pairs V, H below, V is a vector space over $\mathbb{R}$ and H is a subset of V . Select all pairs in which H is a subspace of V when using the same field of scalars and same vector operations.

Syntax advice. For multiple selection questions you are advised to only select answers you are sure are correct because the mark is calculated as the maximum of 0 and

$$\frac{(\text{number of correct selections}) - (\text{number of incorrect selections})}{\text{number of correct answers}}.$$

☐ $V = \mathbb{R}^2$, $H = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} : x, y \in \mathbb{Z} \right\}$

☐ $V = \mathbb{R}^2$, $H = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : xy \geq 0 \right\}$

☐ $V = \mathbb{P}_7$, $H = \{p \in \mathbb{P}_7 : \text{the degree of } p \text{ is } 7\}$

☐ $V = \text{integrable functions on } [5, 8]$, $H = \left\{ f \in V : \int_5^8 f(x)dx = 0 \right\}$

☐ $V = \mathbb{P}_{61}$, $H = \{p \in \mathbb{P}_{61} : p(0) = 29\}$

☐ $V = \mathbb{P}_{94}$, $H = \{p \in \mathbb{P}_{94} : p(x) \leq 0 \text{ for all } x \in \mathbb{R}\}$

☐ $V = \mathbb{P}_{93}$, $H = \{p \in \mathbb{P}_{93} : p(29) = 0\}$

☐ $V = M_{2,6}(\mathbb{R})$, $H = \left\{ \begin{pmatrix} c & a & 0 & a & 0 & a \\ a & c & b & 0 & c & a \end{pmatrix} : a, b, c \in \mathbb{R} \right\}$

☐ $V = \mathbb{R}^2$, $H = \text{span} \left(\begin{pmatrix} -6 \\ 7 \end{pmatrix} \right)$

☐ $V = M_{2,2}(\mathbb{R})$, $H = \{A \in M_{2,2}(\mathbb{R}) : A^T = -A\}$

A past exam question, more writing space

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☐ $V = \mathbb{P}_{94}$, $H = \{p \in \mathbb{P}_{94} : p(x) \leq 0 \text{ for all } x \in \mathbb{R}\}$

☐ $V = \mathbb{P}_{93}$, $H = \{p \in \mathbb{P}_{93} : p(29) = 0\}$

☐ $V = M_{2,6}(\mathbb{R})$, $H = \left\{ \begin{pmatrix} c & a & 0 & a & 0 & a \\ a & c & b & 0 & c & a \end{pmatrix} : a, b, c \in \mathbb{R} \right\}$

☐ $V = \mathbb{R}^2$, $H = \text{span} \left(\begin{pmatrix} -6 \\ 7 \end{pmatrix} \right)$

☐ $V = M_{2,2}(\mathbb{R})$, $H = \{A \in M_{2,2}(\mathbb{R}) : A^T = -A\}$